

i) Challenges in Writing the Program

I had no issues with writing the initial parts of the program. Everything from setting up the fitness scoring, selection, crossover, and mutation was mostly straightforward. My main challenge came up later when testing the code I ran into a fitness plateau at 1.7. This led to many attempts at adjusting the code to increase the maximum fitness score, but none succeeded consistently.

ii) Evaluation of the Generated Schedule

The schedule produced by the program meets the assignment requirements well as far as I can tell. It adheres to the activity, room, and facilitator guidelines provided in the appendix and it successfully applies penalties and rewards based on the details given. The schedule doesn't seem to have any glaring issues or inconsistencies even though the fitness plateau remains.

iii) Potential Improvements

To further improve the program, I could modify the fitness function to create more distinct gradients in fitness scoring. Changing the penalties and rewards for critical factors like room capacity and facilitator load could help distinguish higher-fit individuals. Another possible implementation could be adding more fitness adjustments for "near matches" for example to hopefully encourage more improvements.

iv) Reflections

I used ChatGPT as a resource throughout the development process, mainly to just organize the code and try to troubleshoot the fitness plateau issue. Although it suggested techniques like implementing elitism and adjusting the mutation rate dynamically to promote diversity, none of these methods ended up working. Even though it wasn't able to accurately troubleshoot my issue, it was still helpful in providing a systematic approach to building the program.